Table A1: LNSDE performance on IMTS forecasting

Methods	Human Activity (3000ms → 1000ms)		USHCN (24months → 1month)		PhysioNet (24h → 24h)		MIMIC-III (24h → 24h)	
	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-1}$	MAE$\times 10^{-1}$	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-2}$	MAE$\times 10^{-2}$
LNSDE	3.06 ± 0.05	3.47 ± 0.04	5.05 ± 0.12	3.20 ± 0.12	5.22 ± 0.70	4.17 ± 0.34	1.94 ± 0.54	7.76 ± 0.10

Table A2: LNSDE performance on event prediction tasks

Model	Metric	Synthetic	Neonate	Traffic	MIMIC	StackOverflow	BookOrder
RMTPP	LL (↑)	-0.992 ± .030	-2.231 ± .023	-0.652 ± .012	-1.185 ± .023	-2.352 ± .001	-1.151 ± .007
	Accuracy (↑)	0.839 ± .000	–	0.778 ± .002	0.767 ± .004	0.470 ± .000	0.500 ± .000
	RMSE (↓)	0.322 ± .014	4.614 ± .023	0.363 ± .001	0.822± .017	0.949 ± .000	3.375 ± .003

Table A3: training/inference latency and GPU memory consumption of modles.(ontiFormer(ODE/LNSDE for Q,K,V) means using three different ODEs/LNSDEs to define Q,K,V)

Model	tPacthGNN	VIMTS	ContiFormer	ContiFormer (ODE for Q,K,V)	ContiFormer (LNSDE for Q,K,V)	RosFormer
training latency (s)	7.00	85	69.69	14.32	25.32	8.55
inference latency (s)	3.85	18.82	26.27	5.35	9.26	3.76
GPU memory consumption (MB)	1912	4342	2742	1724	5828	4048

Table A4: Combination Ablation study on IMTS forecasting.

Variant	PhysioNet		Human Activity	
	MSE	MAE	MSE	MAE
RosFormer (Full)	**4.74**	**3.46**	**2.54**	**3.05**
w/o Stochasticity (ODE) and w/o ST-Embedding	5.43	4.00	2.97	3.50
w/o Stochasticity (ODE) and w/o Continuous Attn	4.84	3.65	2.63	3.15
w/o ST-Embedding and w/o Continuous Attn	5.08	3.96	2.66	3.17

Table A5: LNSDE with constant σ diffusion term performance on IMTS forecasting

Methods	Human Activity (3000ms → 1000ms)		USHCN (24months → 1month)		PhysioNet (24h → 24h)		MIMIC-III (24h → 24h)	
	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-1}$	MAE$\times 10^{-1}$	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-2}$	MAE$\times 10^{-2}$
LNSDE with σ	2.76 ± 0.05	3.26 ± 0.04	5.18 ± 0.12	3.16 ± 0.12	4.96 ± 0.70	3.83 ± 0.34	1.82 ± 0.54	7.72 ± 0.10
Latent-ODE	3.32 ± 0.05	3.91 ± 0.04	5.16 ± 0.12	3.21 ± 0.12	6.85 ± 0.70	4.77 ± 0.34	2.11 ± 0.54	7.76 ± 0.10
RosFormer	2.54 ± 0.02	3.05 ± 0.02	4.84 ± 0.03	2.92 ± 0.04	4.74 ± 0.03	3.46 ± 0.05	1.30 ± 0.02	6.32 ± 0.15

Table A6: Other forecasting Baselines performance on IMTS forecasting

Methods	USHCN (24months → 1month)		MIMIC-III (24h → 24h)	
	MSE$\times 10^{-1}$	MAE$\times 10^{-1}$	MSE$\times 10^{-2}$	MAE$\times 10^{-2}$
Grafiti	5.07 ± 0.12	2.97 ± 0.12	1.76 ± 0.12	7.28 ± 0.12
Hyperimts	5.06 ± 0.05	3.29 ± 0.04	1.45 ± 0.12	6.57 ± 0.12
RosFormer	4.84 ± 0.03	2.92 ± 0.04	1.30 ± 0.02	6.32 ± 0.15

Table A7: Other forecasting Baselines performance on event prediction tasks

Model	Metric	Synthetic	Neonate	Traffic	MIMIC	StackOverflow	BookOrder
THP	LL (↑)	-0.589 ± .030	-2.702 ± .023	-0.569 ± .012	-1.137 ± .023	-2.354 ± .001	-1.102 ± .007
	Accuracy (↑)	0.841 ± .000	–	0.818 ± .002	0.834 ± .004	0.468 ± .000	0.622 ± .000
	RMSE (↓)	0.205 ± .014	9.471 ± .023	0.332 ± .001	0.843± .017	0.951 ± .000	3.688 ± .003
RosFormer	LL (↑)	-0.481 ± .020	-2.305 ± .022	-0.510± .015	-0.846 ± .018	-2.314 ± .002	-0.892 ± .008
	Accuracy (↑)	0.835 ± .001	–	0.832± .002	0.853 ± .005	0.497 ± .001	0.636 ± .002
	RMSE (↓)	0.180 ± .004	4.703 ± .030	0.315 ± .002	0.825 ± .006	0.935 ± .001	3.158 ± .015

Table A8: Alation studies

Methods	USHCN (24months → 1month)		MIMIC-III (24h → 24h)	
	MSE$\times 10^{-1}$	MAE$\times 10^{-1}$	MSE$\times 10^{-2}$	MAE$\times 10^{-2}$
RosFormer (Full)	**4.84**	**2.92**	<u>1.30</u>	<u>6.32</u>
w/o Stochasticity	4.96	3.16	1.81	7.75
w/o Linear Const (Gen. SDE)	4.90	3.04	1.52	6.84
w/o GCN Module	4.92	3.04	1.76	7.35
w/o Continuous Attn	4.96	3.12	1.64	7.08
w/o Shared Dynamics	<u>4.86</u>	<u>2.93</u>	**1.26**	**6.31**